

TABLICA LAPLACE-ovih TRANSFORMACIJA      $F(s) = \int_0^{\infty} e^{-st} f(t) dt$

|                                           |                                                                  |
|-------------------------------------------|------------------------------------------------------------------|
| $f(t)$                                    | $F(s)$                                                           |
| $af_1(t) \pm bf_2(t) \quad (a,b) - const$ | $aF_1(s) \pm bF_2(s)$                                            |
| $af(at)$                                  | $aF(s/a)$                                                        |
| $f^{(n)}(t)$                              | $s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0)$ |
| $-tf(t)$                                  | $F'(s)$                                                          |
| $t^2 f(t)$                                | $F''(s)$                                                         |
| $(-1)^n t^n f(t)$                         | $F^{(n)}(s)$                                                     |
| $\int_0^t f(u) du$                        | $\frac{1}{s} \cdot F(s)$                                         |
| $\int_0^t f(u)g(t-u)du = f(t) * g(t)$     | $F(s) \cdot G(s)$                                                |
| $\delta(t)$                               | 1                                                                |
| $h(t)$                                    | $\frac{1}{s}$                                                    |
| $t^n h(t)$                                | $\frac{n!}{s^{n+1}}, \quad n = 1, 2, \dots$                      |
| $t^\nu h(t)$                              | $\frac{\Gamma(\nu+1)}{s^{\nu+1}}, \quad \nu > -1$                |
| $e^{at} h(t)$                             | $\frac{1}{s-a}$                                                  |

|                       |                                                       |
|-----------------------|-------------------------------------------------------|
| $t^n e^{at} h(t)$     | $\frac{n!}{(s-a)^{n+1}}, \quad n = 1, 2, \dots$       |
| $t^\nu e^{at} h(t)$   | $\frac{\Gamma(\nu+1)}{(s-a)^{\nu+1}}, \quad \nu > -1$ |
| $\sin(at)h(t)$        | $\frac{a}{s^2 + a^2}$                                 |
| $\cos(at)h(t)$        | $\frac{s}{s^2 + a^2}$                                 |
| $e^{bt} \sin(at)h(t)$ | $\frac{a}{(s-b)^2 + a^2}$                             |
| $e^{bt} \cos(at)h(t)$ | $\frac{s-b}{(s-b)^2 + a^2}$                           |
| $sh(at)h(t)$          | $\frac{a}{s^2 - a^2}$                                 |
| $ch(at)h(t)$          | $\frac{s}{s^2 - a^2}$                                 |
| $t \sin(at)h(t)$      | $\frac{2as}{(s^2 + a^2)^2}$                           |
| $t \cos(at)h(t)$      | $\frac{s^2 - a^2}{(s^2 + a^2)^2}$                     |

TABLICA KONVOLUCIJA

|                                 |                                                         |
|---------------------------------|---------------------------------------------------------|
| $\cos \omega t * \cos \omega t$ | $(1/(2\omega))(\sin \omega t + \omega t \cos \omega t)$ |
| $\cos \omega t * \sin \omega t$ | $(t/2) \sin \omega t$                                   |
| $\sin \omega t * \sin \omega t$ | $(1/(2\omega))(\sin \omega t - \omega t \cos \omega t)$ |